SYNTHETIC SAMPLE PAPER - fictional, for pipeline testing only

Back-Translation Augmentation for Low-Resource Machine Translation

Abstract

We address low-resource machine translation for Swahili using back-translation with mBART. We evaluate on OPUS-100 and FLORES-200 with BLEU and chrF.

1 Introduction

Back-translation creates synthetic parallel data from monolingual text. Prior work reports gains for several languages.

2 Method

We generate synthetic Swahili data with a reverse mBART model. The synthetic data is mixed with the original parallel data for fine-tuning.

3 Results

mBART alone reaches 25.6 BLEU on FLORES-200 for Swahili. Back-translation reaches 27.0 BLEU on OPUS-100 for Swahili. Back-translation also reaches 51.3 chrF on OPUS-100 for Swahili.

4 Limitations

The quality of the synthetic data depends on the reverse translation model.

5 Future Work

We plan to investigate back-translation for domain adaptation in specialised text.